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Question 1

1. (a) Estimating β_1 with Correlated α_i

When α_i (the village-specific unobserved effect) is **correlated** with X_{it} (the indicator for participation in the fertilizer program), using Ordinary Least Squares (OLS) would lead to biased and inconsistent estimates of β_1 due to this endogeneity.

To address this issue, the researcher should employ the **Fixed Effects (FE) estimator**. The Fixed Effects approach controls for all time-invariant characteristics of the villages, including α_i, by effectively "demeaning" the data or using dummy variables for each village. This transformation eliminates the bias arising from the correlation between α_i and X_{it}, allowing for a consistent estimation of β_1.

Steps to Apply Fixed Effects Estimation:

1. **Within Transformation (Demeaning):**

$$\tilde{Y}_{it} = Y_{it} - \overline{Y}_i$$

$$\tilde{X}_{it} = X_{it} - \overline{X}_i$$

   where $\overline{Y}_i$ and $\overline{X}_i$ are the village-specific averages over time.
2. **Regression on Transformed Data:**

$$\tilde{Y}_{it} = \beta_1 \tilde{X}_{it} + \tilde{u}_{it}$$

   This removes α_i from the equation, isolating the effect of X_{it} on Y_{it}.
3. **Estimation:**
   Perform OLS regression on the transformed (demeaned) data to obtain β_1.

1. (b) Factor Modeled by α_i

α_i captures **time-invariant, unobserved characteristics** of each village that affect crop yield. An example of such a factor is:

- **Soil Quality:** Differences in soil fertility and composition across villages can significantly influence crop yields. Since soil quality does not typically change over the short panel period (2016-2019), it is captured by the village-specific effect α_i.

Other potential factors include:

- Geographic location (e.g., altitude, climate conditions)
- Traditional farming practices

- Access to water resources
- Village infrastructure quality

These unobserved characteristics are critical to control for to avoid omitted variable bias in estimating the effect of the fertilizer policy.

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2. (a) Estimating β_1 with Correlated γ_t

In the specified model:

$$Y_{it} = \beta_1 X_{it} + \gamma_t + u_{it}$$

γ_t represents time-specific unobserved effects that are **common to all villages** in a given year t and are **correlated** with the treatment indicator X_{it}. This correlation introduces endogeneity, meaning that **Ordinary Least Squares (OLS)** estimation would yield biased and inconsistent estimates of β_1.

To obtain a consistent estimate of β_1 in the presence of correlated γ_t, the researcher should **control for time fixed effects**. This can be achieved by incorporating **year-specific dummy variables** into the regression model. Here's how to proceed:

1. **Include Time Fixed Effects:**
   Add a set of dummy variables for each year in the dataset (e.g., 2016, 2017, 2018, 2019). These dummies effectively capture the γ_t terms.

   The extended model becomes:

   $$Y_{it} = \beta_1 X_{it} + \sum_{t=2016}^{2019} \delta_t \cdot \text{Year}_t + u_{it}$$

   where δ_t represents the effect of year t.
2. **Estimation Techniques:**
   - **Fixed Effects (FE) Model with Time Dummies:** Use a fixed effects estimator that includes the time dummies. This controls for any unobserved, time-specific factors (γ_t) that could be influencing both Y_{it} and X_{it}.
   - **Within Transformation:** Alternatively, apply a within transformation that demeans the data by time periods, effectively removing the influence of γ_t.
3. **Run the Regression:**
   Perform an **OLS regression** on the augmented model with time fixed effects. By controlling for γ_t, the estimate of β_1 will capture the **causal effect of the fertilizer program** on crop yields, free from the bias introduced by time-specific unobserved factors.

Key Points:

- **Time Fixed Effects:** Control for any unobserved variables that vary over time but are constant across villages.

- **Consistent Estimation:** By accounting for γ_t, the researcher ensures that the estimation of β_1 is not confounded by time-related shocks or trends that also influence participation in the fertilizer program.

2. (b) Factor Modeled by γ_t

γ_t captures **time-specific, unobserved factors** that affect all villages equally in a given year. Since γ_t is correlated with X_{it}, it likely represents a factor that both influences the decision to participate in the fertilizer program and affects crop yields.

Example Factor: National Weather Patterns

- Weather conditions such as rainfall, temperature, and occurrence of extreme events (e.g., droughts, floods) can vary significantly from year to year.
- **Impact on Crop Yields:** Adverse or favorable weather conditions directly affect crop productivity across all villages. For instance, a drought year would likely reduce crop yields uniformly.
- **Correlation with X_{it}:** In response to unfavorable weather conditions, the government may increase the distribution of free fertilizer to mitigate potential losses, leading to higher participation rates (X_{it}) in those years.

Other Potential Factors Include:

- **National Agricultural Policies:** Introduction of subsidies, price controls, or new farming regulations in a particular year can influence both crop yields and participation in programs.
- **Economic Shocks:** Sudden economic events (e.g., a financial crisis) that affect rural economies might lead to increased reliance on government programs like free fertilizer distribution.
- **Technological Advancements:** Year-specific advancements in agricultural technology or techniques that improve crop yields across all villages.

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3. Advantages of the Model with Both α_i and γ_t

The researcher is now considering the following panel data model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \gamma_t + u_{it}$$

where:

- α_i captures **individual-specific** (village-specific) unobserved effects.
- γ_t captures **time-specific** (year-specific) unobserved effects.
- Both α_i and γ_t are included simultaneously in the model.

Advantages Compared to Models with Only α_i or Only γ_t

1. **Comprehensive Control for Unobserved Heterogeneity:**

- **Previous Models:**
  - **Model 1:** Included α_i (individual fixed effects) but **did not** account for time-specific effects.
  - **Model 2:** Included γ_t (time fixed effects) but **did not** account for individual-specific effects.
- **Current Model:** By including both α_i and γ_t, the model controls for **both** unobserved, time-invariant characteristics of villages **and** unobserved, time-specific shocks that affect all villages. This dual control mitigates potential omitted variable biases arising from either source.

2. **Reduced Omitted Variable Bias:**
   - By accounting for both individual and time effects, the model reduces the risk that omitted variables related to either dimension bias the estimate of β_1. For instance:
     - **Individual Effects (α_i):** Capture inherent village characteristics like soil quality, tradition, or infrastructure.
     - **Time Effects (γ_t):** Capture factors like yearly weather patterns, national policy changes, or economic conditions.
3. **Enhanced Model Specification:**
   - Including both types of fixed effects leads to a more **nuanced and accurate** specification of the data-generating process. It acknowledges that both intrinsic characteristics of villages and external temporal factors can influence crop yields and participation in the fertilizer program.
4. **Improved Estimation Precision:**
   - By controlling for more sources of variation, the model can better isolate the **causal effect** of X_{it} on Y_{it}. This leads to more **precise and reliable estimates** of β_1.
5. **Flexibility in Addressing Multiple Sources of Endogeneity:**
   - If both α_i and γ_t are correlated with X_{it}, the combined model effectively addresses endogeneity arising from **both** individual and time dimensions, ensuring that β_1 is estimated consistently.
6. **Applicability to Complex Real-World Scenarios:**
   - In many practical situations, outcomes are influenced by a combination of individual traits and overarching temporal trends. This model is well-suited to capture such complexities, making it more applicable to real-world data where both sources of unobserved heterogeneity are present.

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4. Order Invariance of Two-Way Demeaning in a Balanced Panel

Understanding the Two-Way Demeaning Process

In the context of the model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \gamma_t + u_{it}$$

two-way demeaning involves removing both entity-specific (α_i) and time-specific (γ_t) fixed effects to isolate the effect of X_{it} on Y_{it}.

Two Approaches:

1. **Entity-First Then Time:**
   - **Step 1:** Subtract the entity mean:

$$Y_{it} - \overline{Y}_i = \beta_1(X_{it} - \overline{X}_i) + (\gamma_t - \overline{\gamma}) + (u_{it} - \overline{u}_i)$$

   - **Step 2:** Subtract the time mean from the result of Step 1:

$$\left(Y_{it} - \overline{Y}_i\right) - \overline{(Y_{i\cdot} - \overline{Y}_i)} = \beta_1(X_{it} - \overline{X}_i) - \overline{X}_i + \gamma_t - \overline{\gamma} + u_{it} - \overline{u}_i - \overline{(\gamma_t - \overline{\gamma})} - \overline{(u_{it} - \overline{u}_i)}$$

     Given a balanced panel, certain averages simplify, leading to the final demeaned equation.
2. **Time-First Then Entity:**
   - **Step 1:** Subtract the time mean:

$$Y_{it} - \overline{Y}_t = \beta_1(X_{it} - \overline{X}_t) + (\alpha_i - \overline{\alpha}) + (u_{it} - \overline{u}_t)$$

   - **Step 2:** Subtract the entity mean from the result of Step 1:

$$\left(Y_{it} - \overline{Y}_t\right) - \overline{(Y_{\cdot t} - \overline{Y}_t)} = \beta_1(X_{it} - \overline{X}_t) - \overline{(X_{it} - \overline{X}_t)} + \alpha_i - \overline{\alpha} + u_{it} - \overline{u}_t - \overline{(\alpha_i - \overline{\alpha})} - \overline{(u_{it} - \overline{u}_t)}$$

     Similarly, leveraging the balanced nature of the panel, the equation simplifies appropriately.

Mathematical Proof of Order Invariance

Let's formalize the two approaches to confirm that the final demeaned variables are identical regardless of the order of demeaning.

Assumptions:

- **Balanced Panel:** Every entity i is observed in every time period t.
- **Definitions:**
  - $\overline{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}$
  - $\overline{Y}_t = \frac{1}{N}\sum_{i=1}^{N} Y_{it}$
  - $\overline{\overline{Y}} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} Y_{it}$

Step-by-Step Proof:

1. **Entity-First Then Time:**
   - **Entity Demeaning:**

$$\tilde{Y}_{it}^{(E)} = Y_{it} - \overline{Y}_i$$

   - **Time Demeaning on Entity-Demeaned Data:**

$$\tilde{Y}_{it}^{(E,T)} = \tilde{Y}_{it}^{(E)} - \overline{\tilde{Y}}_t^{(E)} = Y_{it} - \overline{Y}_i - \left(\overline{Y}_t - \overline{\overline{Y}}\right)$$

$$= Y_{it} - \overline{Y}_i - \overline{Y}_t + \overline{\overline{Y}}$$

2. **Time-First Then Entity:**

- **Time Demeaning:**

$$\tilde{Y}_{it}^{(T)} = Y_{it} - \overline{Y}_t$$

- **Entity Demeaning on Time-Demeaned Data:**

$$\tilde{Y}_{it}^{(T,E)} = \tilde{Y}_{it}^{(T)} - \overline{\tilde{Y}}_i^{(T)} = Y_{it} - \overline{Y}_t - \left(\overline{Y}_i - \overline{\overline{Y}}\right)$$

$$= Y_{it} - \overline{Y}_i - \overline{Y}_t + \overline{\overline{Y}}$$

3. **Comparison:**

$$\tilde{Y}_{it}^{(E,T)} = Y_{it} - \overline{Y}_i - \overline{Y}_t + \overline{\overline{Y}} = \tilde{Y}_{it}^{(T,E)}$$

Therefore, the order in which entity and time demeaning are performed does not affect the final demeaned variable.

Intuitive Explanation

Symmetry in a Balanced Panel:

- In a **balanced panel**, each entity i is observed in every time period t. This uniformity ensures that when you perform entity demeaning (removing entity averages) or time demeaning (removing time averages), the subsequent demeaning step has consistent information to work with across the panel.

Impact of Demeaning Order:

1. **Entity-First Approach:**
   - **First Demeaning:** Removes all entity-specific characteristics by subtracting the entity mean. This isolates within-entity variations over time.
   - **Second Demeaning:** Removes any remaining time-specific effects by subtracting the time mean of the already entity-demeaned data.
2. **Time-First Approach:**
   - **First Demeaning:** Removes all time-specific characteristics by subtracting the time mean. This isolates within-time variations across entities.
   - **Second Demeaning:** Removes any remaining entity-specific effects by subtracting the entity mean of the already time-demeaned data.

Why Order Doesn't Matter:

- **Balanced Nature:** Since every entity appears in every time period, subtracting entity means does not distort the time means, and vice versa. The removal of one set of fixed effects leaves the other set consistent and unaffected.
- **Linear Operations:** Demeaning is a linear operation. When applied sequentially, the order of subtraction of two different sets of means (entities and times) yields the same net result due to the commutative property of addition and subtraction in linear algebra:

$$(Y_{it} - \overline{Y}_i) - \overline{Y}'_t = Y_{it} - \overline{Y}_i - \overline{Y}'_t = (Y_{it} - \overline{Y}'_t) - \overline{Y}_i$$

where $\overline{Y}'_t$ and $\overline{Y}'_i$ are appropriately adjusted means after the first demeaning.

- **Double Centering:** Ultimately, two-way demeaning in a balanced panel centers the data around both entity and time averages, and due to the balanced structure, the centering operations are independent of the sequence.

Question 2

1

1. Difference-in-Differences Function

$$\hat{\delta}_{DiD} = \left[E(Y_i^1|D_i = 1) - E(Y_i^0|D_i = 1)\right] - \left[E(Y_i^1|D_i = 0) - E(Y_i^0|D_i = 0)\right]$$

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2. Key Assumption

To make the difference-in-differences estimator unbiased, the **Parallel Trends Assumption** is required. This assumption states that:

> Without the policy intervention, the trend in consumption changes should be the same for the intervention and control groups.

In detail:

- Without policy intervention, the trend of consumption change for the intervention group (Gansu Province) should match the trend for the control group (other provinces).

It can also be presented in the following equation:

$$E(Y_i^1 - Y_i^0|D_i = 1) = E(Y_i^1 - Y_i^0|D_i = 0)$$

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3. Explanation

The **difference-in-differences estimate** will **overestimate** the treatment effect.

- The DiD method assumes that differences in consumption changes are fully attributable to policy interventions.
- If households in Gansu Province work harder, leading to an increase in income and consumption, this additional increase (unrelated to the policy intervention) will still be interpreted as an effect of the policy by the DiD estimator.

- **Result:** This leads to an **overestimation** of the policy's impact.

Question 3

1. Why a simple comparison may not be a good estimation for the treatment effect?

A simple comparison of employment between New Jersey (NJ) and Pennsylvania (PA) after the minimum wage policy may not accurately estimate the treatment effect for the following reasons:

1. **Pre-existing Differences (Selection Bias):**
   - NJ and PA may have pre-existing differences in labor markets, such as population demographics, economic conditions, or consumer demand. These factors could influence employment levels independently of the minimum wage policy.
2. **Time-Varying Confounders:**
   - Factors unrelated to the minimum wage change (e.g., macroeconomic trends, seasonal employment patterns, or regional shocks) could simultaneously affect employment in both NJ and PA.
3. **Spillover Effects:**
   - Businesses in PA may adjust their policies or pricing in response to the NJ minimum wage increase, which could obscure the treatment effect when comparing post-policy employment levels.
4. **Lack of Counterfactual Information:**
   - A simple post-policy comparison does not provide information on what employment in NJ would have been without the minimum wage increase. This missing counterfactual is necessary to estimate the treatment effect.

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2. What is the regression model you'd like to estimate?

To estimate the treatment effect, the difference-in-differences (DiD) approach can be used. The regression model is:

$$\text{Empl}_{it} = \beta_0 + \beta_1 \text{Treatment}_t + \beta_2 \text{Group}_i + \beta_3 (\text{Treatment}_t \times \text{Group}_i) + \gamma X_{it} + \epsilon_{it}$$

Where:

- Empl_{it}: Employment in restaurant i at time t.
- Treatment_t: Dummy variable (0 for pre-policy, 1 for post-policy).
- Group_i: Dummy variable (1 for NJ, 0 for PA).
- $\text{Treatment}_t \times \text{Group}_i$: Interaction term indicating the treatment effect.

- X_{it}: Control vector.
- β_3: The DiD estimator, capturing the effect of the minimum wage increase on employment in NJ relative to PA.

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3. What is the key assumption of the estimation method you use?

The key assumption of the difference-in-differences (DiD) method is the **parallel trends assumption**, which states:

In the absence of the minimum wage policy, the employment trends in NJ (treatment group) and PA (control group) would have followed the same trajectory over time.

This means that any differences in employment levels before the policy should remain constant over time without the intervention. If this assumption holds, the interaction term $(\text{Treatment}_t \times \text{Group}_i)$ isolates the causal effect of the minimum wage increase.

$$E\big[Empl_{it}|Group_i = 1, Treatment_t = 0\big] = E\big[Empl_{it}|Group_i = 0, Treatment_t = 0\big]$$

4

```
* Set the working directory where your file is located
cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4"

* Import the CSV file into Stata
import delimited DID_Example.csv, clear

* Replace "NA" with missing values
replace empl = "" if empl == "NA"
replace soda = "" if soda == "NA"
replace fries = "" if fries == "NA"

* Convert string variables to numeric where applicable
destring empl, replace
destring soda, replace
destring fries, replace

* Generate the interaction term for difference-in-differences
gen treatmentgroup = treatment * group

* Run the regression model with control variables
reg empl treatment group treatmentgroup fries
```

```
------------------------------------------------------------------------------------------------------------
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4\Q3_4_3.log
  log type:  text
 opened on:  10 Dec 2024, 16:01:33

. do "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4\Q3_4_3.do"

. * Set the working directory where your file is located
. cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4"
D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4

.
. * Import the CSV file into Stata
. import delimited DID_Example.csv, clear
(encoding automatically selected: ISO-8859-2)
(15 vars, 820 obs)

.
. * Replace "NA" with missing values
. replace empl = "" if empl == "NA"
(18 real changes made)

. replace soda = "" if soda == "NA"
(30 real changes made)

. replace fries = "" if fries == "NA"
(45 real changes made)

.
. * Convert string variables to numeric where applicable
. destring empl, replace
empl: all characters numeric; replaced as double
(18 missing values generated)

. destring soda, replace
soda: all characters numeric; replaced as double
(30 missing values generated)

. destring fries, replace
fries: all characters numeric; replaced as double
(45 missing values generated)

.
. * Generate the interaction term for difference-in-differences
. gen treatmentgroup = treatment * group

.
. * Run the regression model with control variables
. reg empl treatment group treatmentgroup fries

      Source |       SS           df       MS      Number of obs   =       761
-------------+----------------------------------   F(4, 756)       =      3.67
       Model |  954.870661         4  238.717665   Prob > F        =    0.0057
    Residual |  49138.3008       756  64.9977524   R-squared       =    0.0191
-------------+----------------------------------   Adj R-squared   =    0.0139
       Total |  50093.1715       760  65.9120677   Root MSE        =    8.0621

------------------------------------------------------------------------------
          empl | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
     treatment |  -2.823965    1.33166    -2.12   0.034    -5.438155   -.2097754
         group |  -3.599658   1.071276    -3.36   0.001    -5.702688   -1.496629
treatmentgroup |   3.566757   1.481293     2.41   0.016     .6588212    6.474694
         fries |   7.365167   2.908898     2.53   0.012      1.65469    13.07564
         _cons |   4.239513   2.617994     1.62   0.106    -.8998884    9.378914
------------------------------------------------------------------------------

.
end of do-file

. log close
```

```
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A4\Q3_4_3.log
  log type:  text
 closed on:  10 Dec 2024, 16:01:54
-------------------------------------------------------------------------------------------------------
```

- Use $fries$ as control:

$$\beta_1 = -2.823965$$

$$\beta_2 = -3.599658$$

$$\beta_3 = 3.566757$$

$$\therefore DiD = 3.566757$$

- Minimum wage increasing from \$4.25 to \$5.05 increases 3.566757 units of full time employees in NJ fast food restaurants compared with fast food restaurants in PA.

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5. Why this is an unbiased estimation of the treatment effect?

This is an **unbiased estimation** of the treatment effect under the following conditions and assumptions:

1. Difference-in-Differences (DiD) Framework

- The Difference-in-Differences (DiD) approach compares the **change in employment** in New Jersey (NJ, the treatment group) before and after the minimum wage increase to the **change in employment** in Pennsylvania (PA, the control group) over the same period.
- The key assumption here is that, absent the treatment, the changes in employment in NJ and PA would have followed **parallel trends** (see Assumption 3 below).

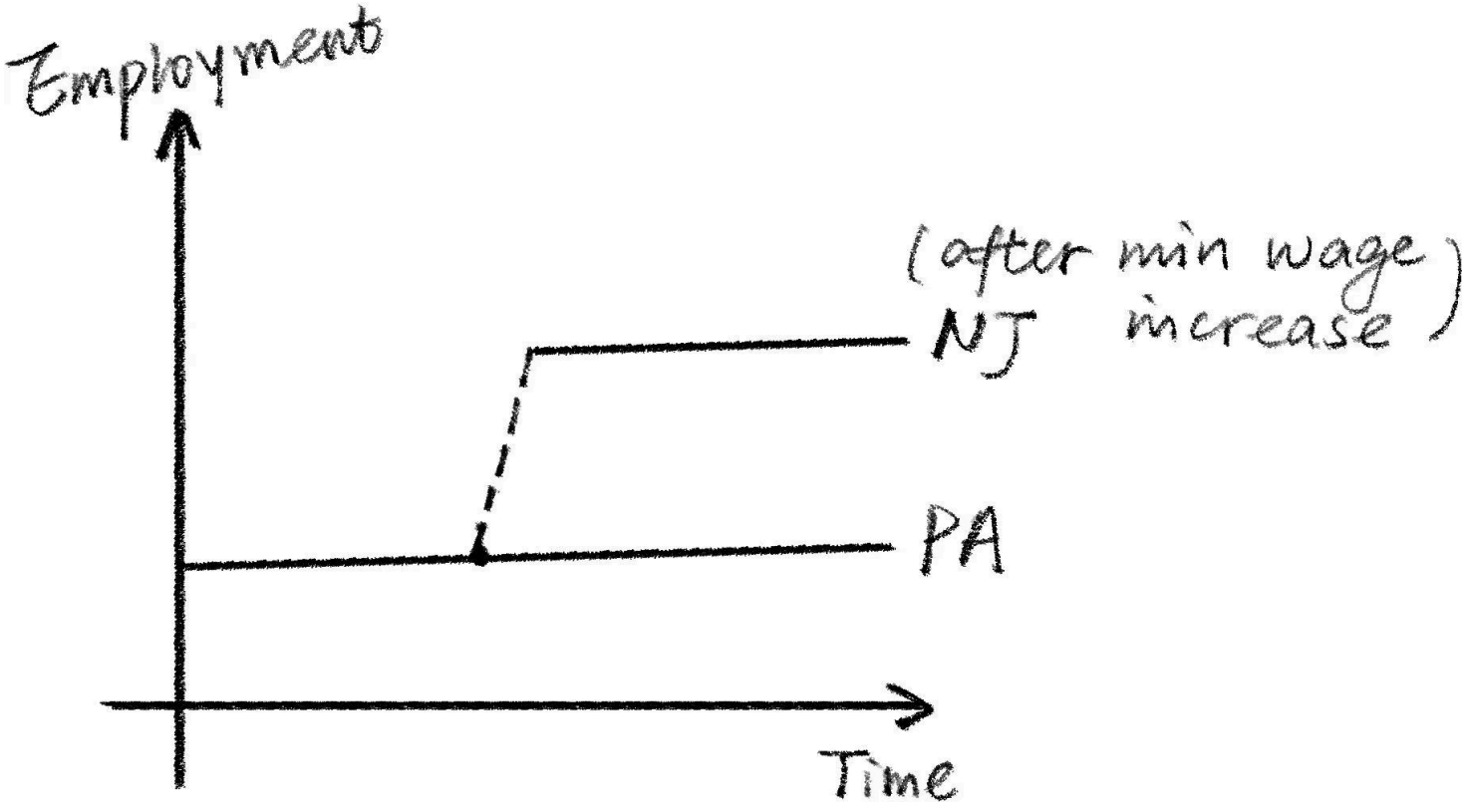
2. Control for Time-Invariant and Time-Varying Factors

- **Group Effect:** The model accounts for fixed differences between NJ and PA (e.g., baseline differences in labor market conditions) through the Group variable.
- **Time Effect:** The model captures changes over time that affect both groups (e.g., macroeconomic trends or seasonal fluctuations) through the Treatment variable.
- By controlling for these factors, the estimated interaction term $\text{Treatment} \times \text{Group}$ isolates the causal impact of the minimum wage increase.

3. Parallel Trends Assumption

- The validity of the DiD estimation relies on the **parallel trends assumption**, which states that in the absence of the treatment, the difference in employment trends between NJ and PA would have remained constant over time.

- If this assumption holds, any deviation from parallel trends post-treatment can be attributed to the minimum wage increase.
- This assumption can be tested by examining pre-treatment trends in the data. If they are parallel, the assumption is more likely to hold.



4. Inclusion of Control Variables

- By including control variables such as:
  - HoursOpening: Reflects the operational intensity of businesses, which may affect employment.
  - Soda and Fries: Prices capturing regional cost or demand factors.
  - COwned: Whether the business is company-owned, which may influence hiring policies.
- These controls reduce potential biases from confounding factors that might correlate with both the treatment and employment.

5. No Spillover Effects

- It is assumed that the policy in NJ did not indirectly affect businesses in PA (e.g., through cross-border worker movement or consumer demand shifts). If spillover effects exist, the control group may no longer represent the counterfactual scenario, leading to bias.

6. No Omitted Variable Bias

- The inclusion of relevant covariates reduces the likelihood of omitted variable bias, ensuring that the treatment effect (Treatment $\times$ Group) is not confounded by unobserved variables.